

Inertia, Leakage, and the Illusion of Progress in Conversational Emotion Forecasting

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Abstract—A de-escalation agent that could tell, one turn in advance, that a customer is about to get angry would be worth building; a model that only *appears* to do so because it peeked at the angry turn itself would not. Emotion-shift forecasting – predicting whether the next utterance will carry a different emotion than the current one, before that utterance is observed – is routinely evaluated with protocols vulnerable to exactly this failure: silent label leakage from the target utterance, compounded by F1 scores that degenerate to a constant “always shift” predictor on imbalanced corpora. We introduce a leakage-safe protocol (explicit commitment point, causal models, a strong speaker $\times$ emotion transition-matrix baseline, AUC + balanced accuracy) and re-evaluate four representative causal model families across six dataset-configurations. On 4 of 6, every safe model loses to the transition matrix and only leaking the future “wins” – yet on one inertia-dominated multimodal case even leaking is not enough, and where safe models genuinely do win, the evidence implicates feature source more than architecture, though this remains a hypothesis pending controlled feature re-extraction. A dose–response study separates two operationally distinct leakage modes with very different inflation profiles. We release the protocol, the baseline, and an automated leakage audit as reusable tools.

Index Terms—emotion forecasting, conversational emotion recognition, evaluation protocol, data leakage, benchmarking

I. INTRODUCTION

Consider a call-center assistant that flags, one turn before it happens, that a customer is about to escalate from neutral to angry – giving the agent seconds to change course. This is the promise of *forecasting* emotion in conversation, as opposed to *recognizing* it: the target state does not yet exist when the prediction is made. It is also precisely the setting in which a model can look right without being right. If any feature – a pooled dialogue embedding, a bidirectional encoder, a context window one utterance too wide – depends even slightly on the turn being forecast, the model is not anticipating the future; it is reading it off a crib sheet built into its own inputs. The failure is silent because nothing in a standard run signals it happened: loss curves look normal, F1 looks respectable, the leaderboard entry looks like progress.

Emotion recognition in conversation (ERC) has a large, mature literature (§II). A smaller but growing thread reframes the task as forecasting – predicting an upcoming emotional shift before the triggering utterance is seen, the setting relevant to anticipatory dialogue systems, de-escalation tools, and turn-taking agents. Forecasting is target-absent by construction: no feature may depend on utterance $n+1$. This constraint is easy

to violate silently, e.g. via bidirectional encoders, dialogue-level pooling, or context windows not truncated at the decision point – and the architectures ERC has spent a decade refining are, almost without exception, built to violate it, because recognition never had to enforce it in the first place.

We set out to ask a narrow measurement question – under a protocol that actually enforces the target-absent constraint, how much of the field’s apparent progress survives – expecting a modest correction: perhaps reported gains would shrink by a few points once leakage was closed. What we found instead was not a uniform discount but a structured picture with real texture: most of the apparent progress on standard corpora is inertia in disguise, the one architecture-vs-baseline comparison that seemed to favor complex models turned out to be entirely a feature-source effect, and leakage itself, once measured on a dial rather than as a single switch, turned out to be at least two mechanistically distinct phenomena with very different inflation profiles. We do not propose a new model. We contribute (i) a leakage-safe protocol and automated audit, (ii) a strong, cheap baseline (a speaker $\times$ emotion transition matrix) that turns out to be surprisingly hard to beat, (iii) a re-evaluation of six causal models – four representative ERC-derived architectures plus two EFC-inspired strategy baselines – across six dataset-configurations, and (iv) a quantification of exactly how much leakage inflates reported numbers, including a dose–response curve that separates those two mechanisms.

II. RELATED WORK

ERC backbones. Our four ERC-derived models are representative of three context-modeling paradigms: recurrent interlocutor-state tracking [7], relational graph convolution [8], [10], and directed relational attention [9]. Knowledge-enriched encoders [5], [11] and multimodal fusion architectures [14]–[18] inform our feature choices (COSMIC RoBERTa; MMDFN [6]-style text+audio+visual). All are, in original form, bidirectional or graph-connected across the full dialogue – appropriate for recognition, but a direct leakage source if repurposed for forecasting without a causal mask; interactive memory [13] and contextual-reasoning networks [12] share this property.

Emotion forecasting (EFC). A parallel line explicitly targets an unobserved future utterance. Altarawneh et al. [20] formalize PEC as predicting turn $n+1$ from context strictly up to n , decomposing the signal into sequence, self-dependency, and recency terms (the strategy we re-implement as PEC-style,

§IV). Shi et al. [21], [22] model bilateral-speaker correlation and word-level cues in text+audio. Hi-EF [19] contributes a dedicated multimodal forecasting benchmark. MREPC [23] pre-trains contrastively over a short-term cross-speaker window. PUGCN [24] substitutes a pseudo-utterance-guided contrastive objective for the absent target (the strategy behind our Pseudo-future-style re-implementation). EDFC [25] targets a next-utterance emotion distribution; ERFC [26] shows feasibility in a deployed call-center setting. All are cross-speaker by design and, to our knowledge, none audits a leakage boundary as strict as ours. Notably EDFC and MM-DFN appeared at ICASSP [6], [25] and the EPC papers at Interspeech [21], [22], making this a speech-community concern as much as an NLP one.

Emotion shift and flip: adjacent, not forecasting. ESD-ERC [27], CFN-ESA [28], and EmoShiftNet [29] use shift labels to sharpen *recognition* of an observed current utterance – computed over turns the model can already see. Emotion-flip reasoning [30], [31] runs backward: given a flip that has already occurred, it attributes the past trigger. Neither predicts an unobserved future shift under a leakage-safe protocol; we exclude both, noting our transition matrix is close in spirit to the inertia signal they are designed to overcome.

Leakage and the illusion of progress. Kapoor and Narayanan [32] survey at least 294 papers across 17 fields and find leakage the dominant cause of irreproducible ML-based findings, proposing the taxonomy our 8-item audit is modeled after. Ferrari Dacrema et al. [33] show most neural recommenders fail to beat tuned classical baselines once evaluation is corrected; Lin [34] makes the same point for neural ranking. We view our transition-matrix result as the same phenomenon in emotion forecasting.

III. PROTOCOL

Commitment point. For decision point n , all input features (text/audio/visual) are restricted to utterances $\leq n$; the target label is $\mathbb{I}[y_{n+1} \neq y_n]$. All six learned models (§IV) are structurally causal, not causal-by-convention.

Causal construction and a zero-tolerance check. DialogueRNN’s global/party/emotion states update by chained GRUCell calls that never touch x_{t+1} . DialogueGCN uses *windowed* causal edges ($j \rightarrow i$ iff $j \leq i$ and $i - j \leq 6$); DAG-ERC uses *unbounded* causal attention ($j \leq i$, any distance) – two families differing sharply in receptive field yet still landing within the same $\lesssim 0.03$ AUC band (§IV), which strengthens rather than weakens Finding 4. We verified causality directly rather than trusting architecture alone: perturbing every input at $t \geq n$ to random noise leaves outputs at $t < n$ changed by exactly 0.0 (floating-point identical) for every model.

Automated leakage audit. Every experiment emits an 8-item checklist: last-utterance masking, model causality, per-utterance feature scope, speaker-split disjointness, val-only threshold tuning, dialogue-level bootstrap, train-fold-only baseline fitting, and baseline completeness.

Strong baseline. Beyond base-rate and no-change/inertia, we fit a speaker $\times$ emotion transition matrix $P(y_{n+1} |$

TABLE I

DATASET STATISTICS AND F1 DEGENERACY. ALWAYS-SHIFT F1 = $2r/(1+r)$ FOR SHIFT RATE r (ANALYTIC; ALWAYS-SHIFT BALANCED ACCURACY IS 0.500 BY CONSTRUCTION, OMITTED). TRANSITION-BA ≈ 0.50 ON MELD/EMORYNLP/IEMOCAP-MM SHOWS STANDARD F1 SITS AT THE DEGENERATE CEILING ON EXACTLY THE CORPORA WHERE SHIFT IS COMMON; ONLY IEMOCAP AND DAILYDIALOG CLEAR IT.

Dataset	#Dial	#Utt	Shift r	Always-shift F1	Transition BA
IEMOCAP	151	7,433	0.429	0.600	0.611
MELD	1,432	13,708	0.581	0.735	0.500*
EmoryNLP	827	9,489	0.722	0.839	0.500*
DailyDialog	13,118	102,979	0.198	0.331	0.681
IEMOCAP-mm	151	7,433	0.429	0.600	0.500*
MELD-mm	1,432	13,708	0.581	0.735	0.500*

y_n , speaker) on the train fold only (Laplace $\alpha=1.0$; a speaker’s row is used only with ≥ 20 training observations, else global backoff). This is the baseline every model must beat.

Hyperparameters. No per-architecture tuning: one fixed configuration (Adam, lr 10^{-3} , batch 32, hidden 128, focal loss, 20 epochs, no early stopping, no LR/hidden-size/epoch search) is applied to all six learned models on every dataset, removing unequal tuning effort as a confound. The only validation-set-selected value is the binary decision threshold (grid search maximizing shift-F1), used only for F1/precision/recall/balanced accuracy; AUC is threshold-free.

Metrics. We report shift-AUC as primary alongside shift-F1, precision/recall, and balanced accuracy, auto-flagging degenerate predictors (prediction rate $>98\%$ or $<2\%$ one class).

Significance. Model-vs-baseline comparisons use a paired, cluster-robust bootstrap (dialogue-level resampling, $n=2000$), Holm–Bonferroni-corrected *jointly* across all 6 models $\times$ 6 datasets (36 comparisons, one family).

IV. EXPERIMENTS

Data. Four text corpora with off-the-shelf RoBERTa features (IEMOCAP, MELD, EmoryNLP, DailyDialog; COSMIC pipeline), plus two multimodal configurations (IEMOCAP, MELD with MM-DFN text+OpenSmile-audio+DenseNet-visual features). Table I gives size and shift-rate statistics (0.20–0.72) together with an F1-degeneracy analysis referenced below.

Main result (Table II). We evaluate six causal models: four ERC-derived architectures – plain GRU and causal re-implementations of DialogueRNN [7], DialogueGCN [8], and DAG-ERC [9] – and, to test whether inertia-dominance is an artifact of repurposing recognition architectures rather than of the task itself, two EFC-inspired strategy baselines: a PEC-style sequence/self-dependency/recency decomposition [20] (our transition matrix is close to a degenerate limit of its self-dependency term) and a Pseudo-future-style predict-then-classify model in the spirit of the pseudo-utterance-guided contrastive strategy [24] (both faithful “-style” re-implementations; original code is unavailable). All six sit within $\lesssim 0.03$ AUC of each other on every dataset (max spread: IEMOCAP, 0.029). Whether any of them beats the transition matrix is decided by the dataset, not the model family: on 4 of 6 configurations – IEMOCAP, MELD, EmoryNLP (text), and IEMOCAP-mm – every safe model, including

both EFC-inspired baselines, loses; only DailyDialog and MELD-multimodal see all six win. The EFC-inspired baselines reproduce the exact same win/loss pattern as the four ERC-derived architectures without any tuning toward this outcome – the paper’s central new evidence that inertia-dominance is not an ERC-repurposing artifact. On three of the four losing configurations, a leaky bidirectional model “recovers” a win by seeing the future; on the fourth (IEMOCAP-mm), even the leaky upper bound (0.590) stays below the transition matrix (0.642), and Pseudo-future-style is in fact a jointly *significant loser* there ($\Delta\text{AUC} = -0.117$; Table II caption) – inertia dominates so completely that no amount of future-peeking, nor an EFC-inspired strategy, is enough.

Scope, and why these are not straw men. Two boundaries should be explicit. First, four of our six models re-implement representative context-modeling *families* rather than specific published EFC systems (unavailable code); the other two are faithful EFC-inspired-style re-implementations. Our claims concern evaluation practice and model families, not any individual published number. Second, one may worry the causal re-implementations are artificially weak. Our results argue otherwise: (i) six independently implemented paradigms – plain recurrence, speaker-state recurrence, a fixed 6-utterance causal window (DialogueGCN), unbounded causal attention (DAG-ERC), and two EFC-inspired strategy baselines – land within $\lesssim 0.03$ AUC of one another on *every* dataset despite sharply different receptive fields, and six coincidental bugs do not fail in lockstep; (ii) causality was verified directly, not assumed: perturbing all $t \geq n$ inputs to noise leaves $t < n$ outputs unchanged to floating-point exactness for every model (§III); (iii) the leaky variant of the same pipeline reaches 0.75–0.82 AUC where safe models collapse, so it demonstrably learns when signal is available; and (iv) no per-architecture tuning was performed, so unequal tuning effort cannot explain the lockstep either.

Notes on Table II numbers. Point-estimate differences will not exactly match paired-test ΔAUC values (Table II averages 3 seeds; the paired bootstrap uses a single seed’s raw predictions, required for well-defined per-utterance pairing) but agree in sign and magnitude. DailyDialog’s numbers use the *original* (pre-deduplication) split; 13.4% of its test dialogues are exact train duplicates, but a robustness check on the duplicate-free split confirms the GRU win survives ($\Delta\text{AUC} +0.045 \rightarrow +0.038$, both $p < .01$).

Feature source, tentatively. The MELD win/loss flips entirely on feature source at fixed architecture: COSMIC RoBERTa text loses ($p < .001$, joint-significant); MM-DFN text+audio+visual wins ($p = .021$ uncorrected, not joint-significant). Table III shows the gain is carried by MM-DFN’s text channel alone. We identified two candidate explanations without adjudicating between them: MM-DFN’s 600-d task-adapted textCNN/GloVe encoding may carry a cleaner shift signal than COSMIC’s 1024-d frozen RoBERTa embedding, *or* the two pipelines’ text preprocessing is simply not verified to be identical. We did not run a controlled re-extraction to distinguish these, so “feature source rather than architecture”

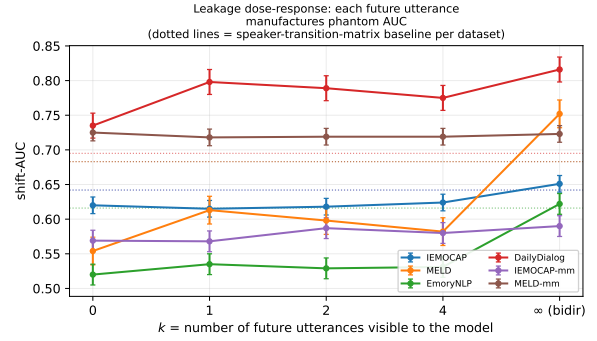


Fig. 1. Leakage dose–response across six dataset-configurations. Dotted lines mark the speaker-transition-matrix baseline per dataset. $k=0 \rightarrow 1$ (seeing the target utterance) drives most of the finite-window inflation; a second jump appears only at full bidirectional context ($k=\infty$).

is reported as a hypothesis, not a conclusion.

Leakage quantification. A bidirectional (“leaky”) GRU inflates AUC by $+0.03$ to $+0.20$ over the safe GRU (largest on MELD). A capacity-matched causal control (same parameter count) shows the effect is not primarily capacity: mean gap shrinks only $0.105 \rightarrow 0.093$ under capacity control.

Dose–response (Fig. 1). We sweep $k \in \{0, 1, 2, 4, \infty\}$ future utterances visible to the model (mean-pooled window for finite k , full bidirectional at ∞). The curve is *not* a smooth ramp: on MELD/DailyDialog, $k=0 \rightarrow 1$ (seeing the target utterance itself) accounts for most of the windowed inflation (MELD $+0.059$); $k=1 \rightarrow 4$ is flat or slightly declines (averaging in more distant future utterances dilutes the sharp $n+1$ signal); a second, larger jump appears only at $k=\infty$, where each future utterance gets its own backward-pass representation rather than being averaged. On IEMOCAP-mm/MELD-mm the curve is flat across the *entire* range – leakage has nothing to add when the honest signal already suffices. One caveat: because finite- k leakage is mean-pooled while $k=\infty$ is a full bidirectional pass, the contrast conflates *access* to the future with its *representation*; we therefore claim two operationally distinct leakage modes, not a mechanistic decomposition.

V. FINDINGS AND DISCUSSION

Read individually, the six results below are a checklist. Read together, they tell a more specific story than “the field is exaggerating progress.”

- 1) Apparent progress on 3/4 text corpora is largely inertia – a train-fold-only transition matrix is a hard baseline current models do not beat when the future is truly hidden.
- 2) Standard F1 is degenerate on high-shift-rate corpora; AUC + balanced accuracy are required to see it.
- 3) Leakage manufactures up to $+0.20$ AUC, is not primarily a capacity artifact, and is largest exactly where the legitimate signal is weakest.
- 4) Model choice (GRU vs. five other structurally and strategically distinct families, including two EFC-inspired strategy baselines) contributes negligibly ($\lesssim 0.03$ AUC) next to feature choice and leakage.

TABLE II

SHIFT-AUC ACROSS SIX DATASET-CONFIGURATIONS AND SIX CAUSAL MODELS. * = DEGENERATE (AUTO-FLAGGED, PREDICTION RATE >98% ONE CLASS). **BOLD** = POINT ESTIMATE EXCEEDS THE TRANSITION MATRIX. PAIRED, CLUSTER-ROBUST BOOTSTRAP TESTS, HOLM-CORRECTED *jointly* ACROSS THE RESULTING 36 COMPARISONS (6 MODELS $\times$ 6 DATASETS): 10/36 SURVIVE. ALL SIX MODELS’ MELD LOSSES ARE JOINTLY SIGNIFICANT ($p < .0001$ EACH) – THE MOST ROBUST RESULT IN THE PAPER. ON DAILYDIALOG ONLY DAG-ERC’S WIN SURVIVES THIS STRINGENCY ($p=.001$); GRU’S NARROWLY MISSES ($p=.002$, THRESHOLD .0019), AND DIALOGUERNN/DIALOGUEGCN/PEC/PSEUDO-FUTURE’S WINS DO NOT CLEAR IT. PSEUDO-FUTURE’S IEMOCAP-MM LOSS IS JOINTLY SIGNIFICANT ($p=.001$). ALL MELD-MM WINS AND THE REMAINING DAILYDIALOG WINS ARE DIRECTIONALLY CONSISTENT (POSITIVE POINT ESTIMATE, ALL SIX MODELS) BUT NOT JOINTLY SIGNIFICANT AT THIS STRICTER THRESHOLD; WE REPORT THIS PLAINLY RATHER THAN ADOPT A MORE FAVORABLE PER-MODEL CORRECTION. FULL 36-ROW CORRECTION TABLE AND PER-SEED RESULTS ARE IN THE RELEASED ARTIFACTS.

Dataset	Transition	GRU	DRNN	DGCN	DAG-ERC	PEC	PseudoFut	Leaky
IEMOCAP	0.642	0.621	0.613	0.622	0.593	0.628	0.620	0.651
MELD	0.683	0.554*	0.564*	0.574*	0.575*	0.562	0.555	0.752
EmoryNLP	0.616	0.520*	0.528*	0.548*	0.522*	0.556	0.520	0.622
DailyDialog	0.695	0.735	0.721	0.726	0.727	0.734	0.728	0.816
IEMOCAP-mm	0.642	0.569	0.569	0.553	0.557*	0.576	0.563	0.590
MELD-mm	0.683	0.725	0.707	0.724	0.718	0.723	0.723	0.723

TABLE III

MELD FEATURE-SOURCE ABLATION, SAFE GRU, SHIFT-AUC. THE GAIN OVER ROBERTA TEXT IS CARRIED ENTIRELY BY MM-DFN’S TEXT CHANNEL; IN THIS LOGGED ABLATION, AUDIO/VISUAL CHANNELS DO NOT IMPROVE OVER MM-DFN TEXT-ONLY.

Features	AUC (none)
COSMIC RoBERTa text (1024-d)	0.554
MM-DFN text-only (600-d)	0.724
MM-DFN text+audio	0.722
MM-DFN text+visual	0.705
MM-DFN text+audio+visual	0.712

- 5) Leakage is (at least) two operationally distinct modes, not one: target-utterance exposure and full bidirectional context, with very different inflation profiles.
- 6) The two EFC-inspired baselines (PEC-style, Pseudo-future-style) fail in the *identical* pattern as ERC-derived architectures – inertia-dominance is a property of the task and evaluation protocol, not an artifact of repurposing recognition architectures.

The story is not “nothing works.” *Whether* a model beats the transition matrix is decided almost entirely by where its features come from and how much of the future its evaluation quietly admits – not by which of six strategically distinct models processes those features (Findings 4, 6). A field that keeps introducing new context-modeling mechanisms and forecasting strategies while the real lever is feature source and leakage discipline is optimizing the wrong variable. The dose-response curve (Fig. 1, Finding 5) sharpens this: widening a context window by one or two future utterances inflates numbers far less than a fully bidirectional encoder, but the target utterance alone already accounts for most of the damage – the single highest-leverage fix is not a longer checklist, it is masking one utterance correctly.

IEMOCAP-mm resists the story on both sides: even the leaky upper bound, and even a dedicated EFC-inspired baseline, cannot clear the transition matrix – the cleanest sign that inertia is close to the entire signal there, not a residual. Conversely, the two configurations where safe models win on point estimates are not evidence that model choice matters after all: the MELD feature-source comparison (*Feature source*, *tentatively*, §IV) shows the identical GRU losing with one

feature set and winning with another, holding the model fixed. Progress here, where real, lives in representation choice, not the model on top of it.

Limitations. Our speaker-split concern is only partly addressed: we rule out exact-duplicate-dialogue contamination (DailyDialog), but not subtler speaker-identity or writing-style leakage in non-duplicate dialogues (MELD/EmoryNLP/DailyDialog speaker IDs are per-dialogue local, not verified global identities – only IEMOCAP is genuinely speaker-independent); left to future work, as is the feature-source hypothesis pending a controlled re-extraction (§IV).

VI. MINIMAL REPORTING CHECKLIST

Standalone checklist for future EFC-forecasting papers: (1) state the commitment point and verify no feature crosses it; (2) verify causality directly (e.g. a future-perturbation test), not by architecture claim alone; (3) report a train-fold-only transition-matrix baseline; (4) report AUC and balanced accuracy, not F1 alone, and flag degenerate predictors; (5) use a paired, cluster-robust (dialogue-level) significance test, corrected jointly across the full model $\times$ dataset family; (6) label any “leaky” ablation and exclude it from headline comparisons; (7) check train/test for exact-duplicate dialogues before reporting a positive result.

VII. CONCLUSION

We asked how much apparent progress in conversational emotion forecasting survives a protocol that actually hides the future. It depends on where you look: inertia dominates most standard corpora – for ERC-derived architectures and two EFC-inspired strategy baselines alike – rare wins trace to representation rather than model choice, and leakage is itself two operationally distinct modes, not one. None of this required a new model; it required refusing to let the target utterance, or the turns after it, into the input. Similar audits recur across fields [32]–[34], wherever architectures were designed for a different temporal contract than the one they are repurposed to solve. Protocol, baseline, self-checks, and audit are prepared for anonymous release upon submission.

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